

Inferring Structural Causal Models from Data with Grammatical Evolution

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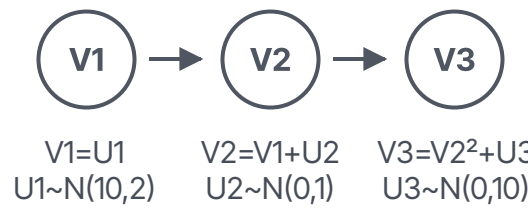
From Causal Models to Probabilistic Programs

SCM (Structural Causal Model)

A tuple (V, U, F) : endogenous variables V , independent exogenous noise U , structural functions F .

$$V_i = f_i(V_{i_1}, \dots, V_{i_h}, U_i)$$

Example: a chain



ANM (Additive Noise Model)

Restricts F so noise enters only additively:

$$V_i = f_i(V_{i_1}, \dots, V_{i_h}) + U_i$$

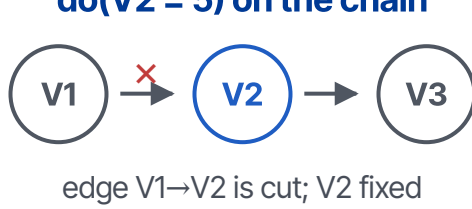
as a SOGA program

```
v_1 = gm(1,10,2)
v_2 = v_1 + gm(1,0,1)
v_3 = v_2 * v_2 + gm(1,0,10)
```

Interventions

Setting a variable to a fixed value — $do(V_i=v)$ — cuts its incoming edges. Everything else stays unchanged, letting us tell **causal structure** apart from mere **correlation**.

Two models can agree on every observed correlation and still disagree on **what causes what**. Interventional data breaks that tie: fitness is evaluated both observationally and under each intervention, rewarding programs that get the **mechanism** right, not just the numbers.



as a SOGA program

```
v_1 = gm(1,10,2)
v_2 = 5
v_3 = v_2 * v_2 + gm(1,0,10)
```

Causal structure is rarely known in advance

A structural causal model explains **cause** and **effect**: each variable is a function of its causes, plus independent noise. In practice we only observe data — the **structure** has to be inferred.

Grammar & Fitness

How a candidate causal model is built, and how it is scored

Grammar

```
{prog} ::= {ass1}...{assn}{perm}
{assi} ::= t1={noise};
{assi} ::= {structi}; t[i]=t[i]+{noise}  Vi,2≤i≤n
{structi} ::= t[i]={funi};  Vi,2≤i≤n
{noise} ::= uniform([s]{r},{r}},{i}) | gm({tripl})
{tripl} ::= {r},{s}{r},{r},{tripl} | {r},{s}{r},{r}
{r} ::= {d}{d}E(s){e}   {e} ::= 0|1|2

{i} ::= 0|1|...|50   {s} ::= +|-
{d} ::= 0|1|...|9
{funi} ::= {depi} | {s}{r} | ({funi}){op}{funi}
Vi,2≤i≤n
{depi} ::= t1 | ... | t[i-1]  Vi,2≤i≤n
{op} ::= +|-|*
{perm} ::= v1=t1;...;v[n]=t[n] | ... (n! rules)
```

Expressible ANMs: exogenous noise is Gaussian-mixture or uniform; structural functions use only +, −, ×.

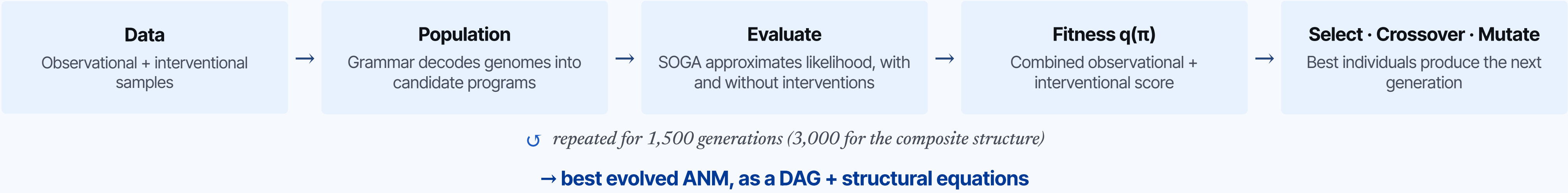
Fitness

A candidate π is scored by log-likelihood: how probable the data is under it — observed, and after intervening.

$$q(\pi) = \log \mathcal{L}(\pi \mid D) + \sum_{(D_j, V_{i_j}, v_j) \in I} \log \mathcal{L}(\pi \mid_{do(V_{i_j}=v_j)} \mid D_j).$$

The evolutionary search

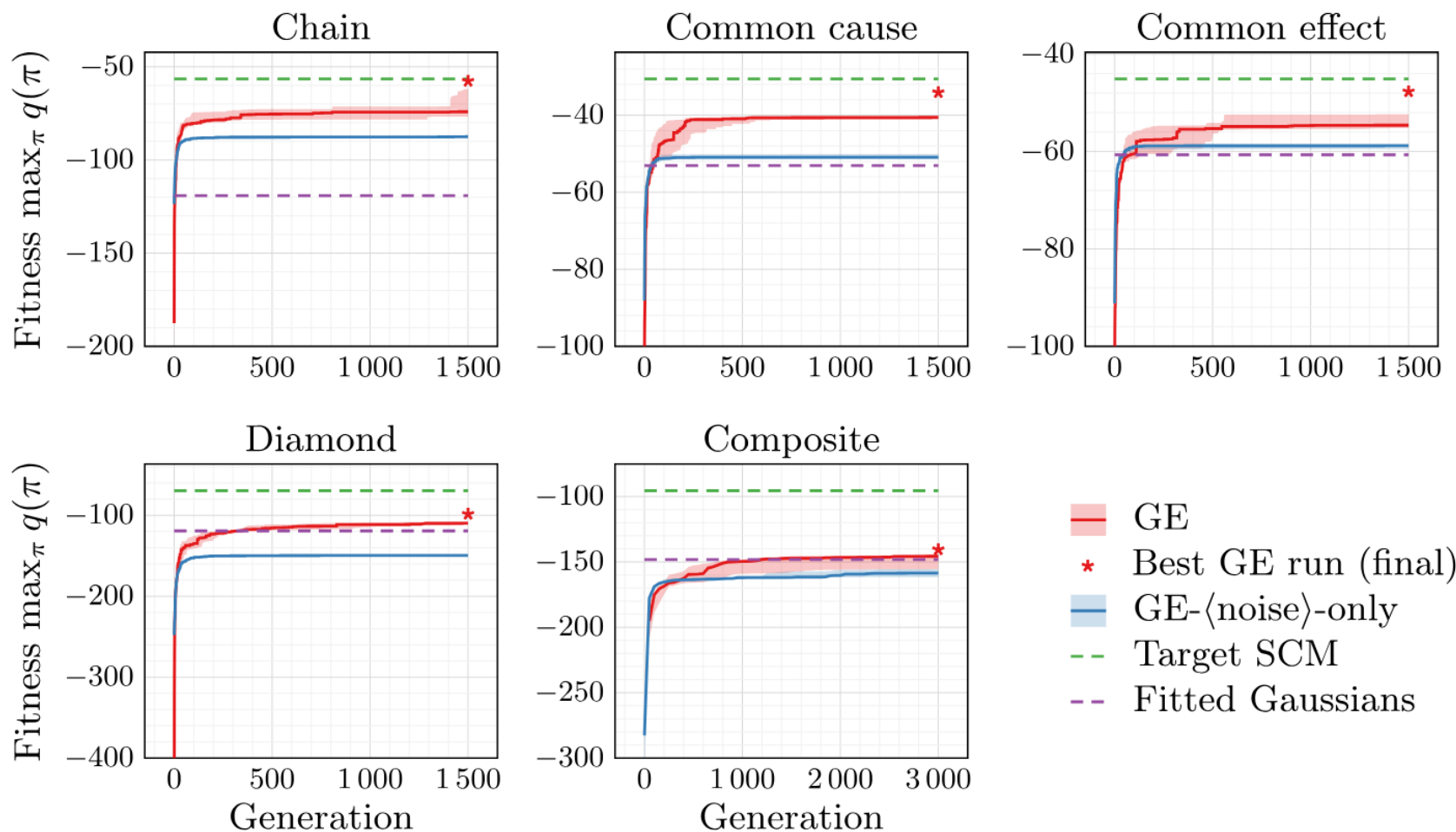
How GE turns data into a causal program



Does it work?

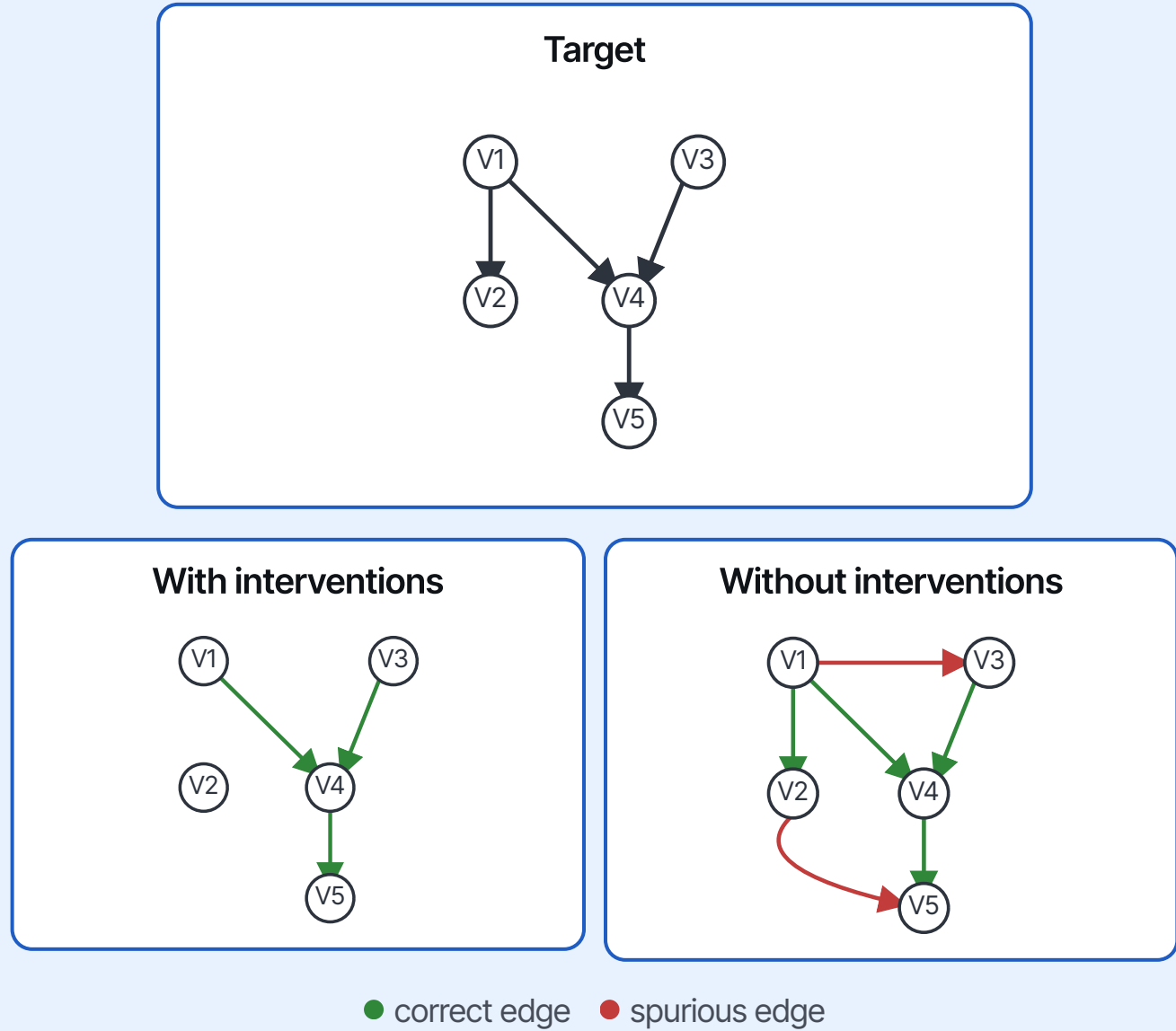
Evaluated on chain, common cause, common effect, diamond, composite

Fitness across generations, all five benchmarks



Modeling dependencies among variables – not just marginals – gets fitness closest to the true model.

Composite problem (5 variables) — interventions shape what's recovered



Without interventions the search recovers more edges – but several point the wrong way.

Evolution can discover not just what fits the data – but what caused it.